**Supplementary Materials**

# Supplementary Tables

## Supplementary Table 1: National Institutes of Health Assessment of Controlled Intervention Studies Scoring

| Study | Q1: Study described as RCT? | Q2: Randomization method? | Q3: Allocation Concealment? | Q4: Participants & Provider Blinded? | Q5: Outcome assessors blinded to group assignment? | Q6: Similar groups at baseline? | Q7: Dropout <20%? | Q8: Differential Dropout between groups <15%? | Q9: Adherence to protocol? | Q10: Absence of other/different interventions between groups? | Q11: Outcome measurements valid/consistent? | Q12: Sufficient n/ power to detect differences in main outcome between groups with 80% power? | Q13: Pre-specified outcomes/subgroup analysis? | Q14: Analyzed in group originally assigned (i.e. ITT)? | Quality |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Antonia, et al. 2018 (1) | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | 14/14 |
| Barlesi, et al. 2018 (2) | Y | Y | N | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 11/14 |
| Borghaei, et al. 2015 (3) | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | Y | NR | 11/14 |
| Brahmer, et al. 2015 (4) | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 12/14 |
| Carbone, et al. 2017 (5) | Y | Y | Y | N | Y | Y | Y | Y | Y | Y | Y | Y | N | NR | 10/14 |
| Fehrenbacher, et al. 2016 (6) | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 12/14 |

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Gandhi, et al. 2018 (7) | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | 14/14 |
| Govindan, et al. 2017 (8) | Y | Y | Y | Y | Y | Y | Y | Y | NR | Y | Y | Y | Y | Y | 13/14 |
| Hellmann, et al. 2019 (9) | Y | Y | Y | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | NR | 12/14 |
| Herbst, et al. 2016 (10) | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 12/14 |
| Jotte, et al. 2020 (11) | Y | Y | NR | N | N | Y | Y | Y | Y | Y | Y | NR | Y | Y | 10/14 |
| Mok, et al. 2019 (12) | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 12/14 |
| Nishio, et al. 2020 (13) | Y | Y | Y | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 12/14 |
| Paz Ares, et al. 2018 (14) | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | 14/14 |
| Reck, et al. 2016 (15) | Y | Y | Y | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | Y | 13/14 |
| Rittmeyer, et al. 2017 (16) | Y | Y | N | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 11/14 |
| Socinski, et al. 2018 (17) | Y | Y | NR | N | N | Y | Y | Y | Y | Y | Y | Y | Y | Y | 11/14 |
| West, et al. 2019 (18) | Y | Y | N | N | N | Y | Y | Y | Y | Y | Y | N | Y | Y | 10/14 |

[a] National Institutes of Health (2021a). Quality Assessment of Controlled Intervention Studies. Available online at: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools (Accessed 4/14/21). Y=Yes, N=No, NR=Not reported, ITT= Intention-to-treat, RCT=Randomized-control trial.

**Supplementary Table 2: National Institutes of Health Assessment for Observational Cohort and Cross-Sectional Studies** [a]

| Study | Q1: Research objective clearly stated? | Q2: Study population specified? | Q3: 50% participation rate among eligible? | Q4: Recruitment from similar populations/ times? | Q5: Justification for n/power/ effect estimate? | Q6: Exposures measured before outcomes being measured? | Q7: Sufficient timeframe? | Q8: Different levels of exposure measured? | Q9: Exposure measures consistent across all participants? | Q10: Exposure assessed more than once? | Q11: Valid/ consistently implemented outcome measures | Q12: Outcome assessor blinded? | Q13: <20% loss to follow up | Q14: Measurement of/adjustment for confounding variables? | Quality |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Adachi, et al. 2019 (19) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Ahn, et al. 2019 (20) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Anouti, et al. 2020 (21) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Chen, et al. 2020 (22) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Elkrief, et al. 2020 (23) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Foster, et al. 2019 (24) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Huang, et al. 2020 (25) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Kano, et al. 2020 (26) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Lichtenstein, et al. 2020 (27) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Lin, et al. 2018 (28) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Nazha, et al. 2020 (29) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Ng, et al. 2018 (30) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Prelaj, et al. 2019 (31) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Smit, et al. 2020 (32) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Song, et al. 2020 (33) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |
| Yang, et al. 2020 (34) | Y | Y | Y | Y | N | Y | Y | NA | Y | Y | Y | NA | NA | Y | 10/14 |

[a] National Institutes of Health (2021b). Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies. Available online at: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools (Accessed 4/14/21). Y=Yes, N=No, NA=Not Applicable, ITT= Intention-to-treat, RCT=Randomized-control trial.

Supplementary Table 3: Association between survival and immunotherapy stratified by race in experimental studies (n=5)

| | | Immunotherapy vs. No Immunotherapy by Race HR (95% CI) | | | | |
|---|---|---|---|---|---|---|
| Clinical Trial | Author, Year | All | Asian | Black | "OtherRaces" | White |
| CA184-104 | Govindan, et al. 2017 (8) | 0.91 (0.77, 1.07) | 0.73 (0.51, 1.03) | 0.99 (0.16, 6.12) | N/A | 0.94 (0.78, 1.15) |
| PACIFIC | Antonia, et al. 2018 (1) | 0.68 (0.47, 1.00) | 0.62 (0.38, 1.01) | N/A | N/A | 0.71 (0.54, 0.93) |
| IMpower 131 | Jotte, et al. 2020 (11) | 0.88 (0.73, 1.05) | 1.31 (0.70, 2.42) | 0.67 (0.12, 3.67) | N/A | 0.84 (0.69, 1.02) |
| IMpower132 | Nishio, et al. 2020 (13) | 0.86 (0.71, 1.06) | 0.73 (0.46, 1.15) | N/A | N/A | 0.90 (0.72, 1.14) |
| JAVELIN Lung 200 | Barlesi, et al. 2018 (2) | 0.90 (0.73, 1.12) | 0.95 (0.56, 1.61)[c] | 0.96 (0.52, 1.77)[b] | 0.89 (0.71, 1.13)[d] | 0.87 (0.68, 1.11)[a] |

[a] For this study, this category was designated Non-Hispanic/Non-Latino. HR = hazard ratio; N/A = not applicable.

[b] For this study, this category was designated Hispanic or Latino.

[c] For this study, this category was designated Japanese Living in Japan.

[d] For this study, “Other Race” includes patients who were not defined in the categories listed in footnotes b-c. No further breakdown of these categories was provided in the study

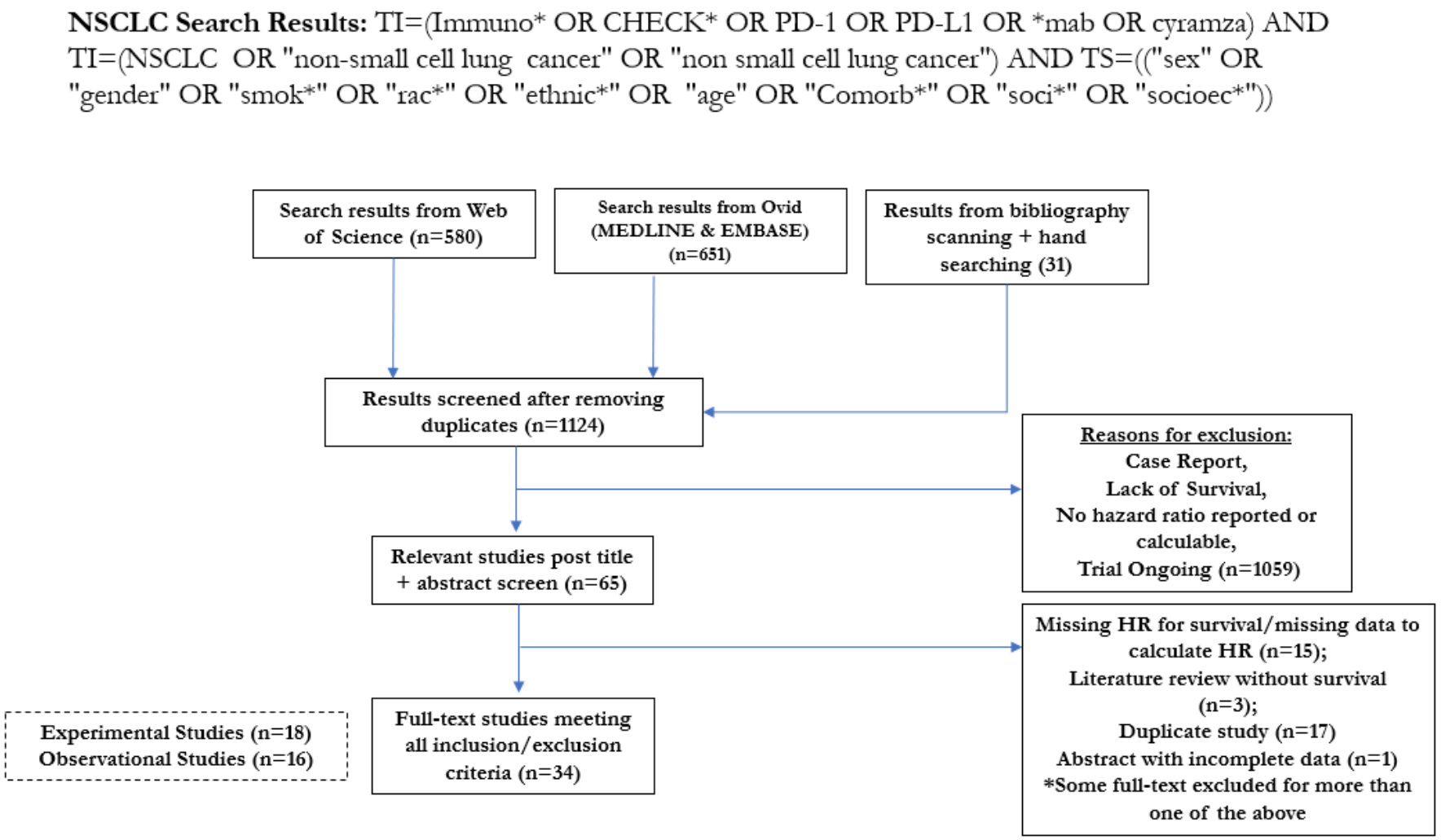
Supplementary Table 4: Association between survival after immunotherapy and race in observational studies (n=3)

| | Race | | | | |
|---|---|---|---|---|---|
| Author, Year | Reference | Comparison | Univariate HR (95% CI) | Mutlivariate HR (95% CI) | Adjustments |
| Foster, et al. 2019 (24) | White | Non-White | N/A | 0.89 (0.82, 0.98) | NR |
| Nazha, et al. 2020 (29) | White/Caucasian | Black/African American | 0.9 (0.59, 1.37) | N/A | N/A |
| | | Other | 0.33 (0.10, 1.04) | | |
| Ng, et al. 2018 (30)[a] | Asian | Non-Asian | 0.60 (0.36, 1.00) | 2.02 (0.49, 8.26) | NR |

[a] Progression free survival (PFS). HR = hazard ratio; N/A = not applicable; NR = not reported

## Supplementary Figure

**Supplementary Figure 1:** PRISMA flowchart of search results. HR = hazard ratio.



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